MultiLabel-Toxicity-Moderation

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Introduction & Objective

Goal

Engineer an automated content moderation engine capable of precisely identifying overlapping categories of online harassment in a single inference pass.

Challenge

Unlike standard classification, this model must determine probabilities for six simultaneous toxicity categories: toxic, severe_toxic, obscene, threat, insult, and identity_hate.and rare classes.

Methodology & Approach

Data Preparation

- Baseline: Text converted using TF-IDF Vectorization to establish word-frequency features
- Advanced: Comments tokenized, padded to 150 words, and processed with dense Word Embeddings

Model Pipeline

- Baselines: Implemented Logistic Regression and Linear SVC (SVM) using the One-vs-Rest strategy

Conclusion

- The lower F1-Scores across all models reflect the extreme class imbalance challenge of rare toxic labels
- However, robust ROC-AUC scores prove the LSTM's exceptional ability to correctly rank toxic comments higher than clean ones
- The Bidirectional LSTM succeeded by understanding the sequence and context of words, capturing sarcasm and complex insults that word-counting methods (TF-IDF) completely miss

CORE ENGINE

Deep Learning Architecture (Bi-LSTM)

Advanced neural network capturing bidirectional context and semantic relationships

Input Layer — Max length 150 → **Embedding Layer** — 20K vocab, 128 dim → **Bidirectional LSTM** — 60 units → **GlobalMax Pool1D** — Feature extraction → **Dropout** — Prevent overfitting → **Dense Layer** — 50 units, ReLU → **Output Layer** — 6 sigmoid neurons

Total Parameters
~2.8 Million

Results & Comparative Analysis

	Logistic Regression (Baseline)	Linear SVC (SVM Baseline)	Bidirectional LSTM (Deep Learning) — Best Model
MACRO F1-SCORE	0.4682	0.6074	0.6358
OVERALL ROC-AUC	0.9080	N/A	0.9766

Key Observation The LSTM model achieved a decisive victory with an overall ROC-AUC of **0.9766**, excelling at detecting rare classes like 'severe_toxic' (~98.5% accuracy).